No.	Question	Answer
1	**What is a Boolean data type in Arduino?**	A Boolean variable stores one of two values: **true or false**. It occupies **one byte** of memory.
2	**What is the use of the `char` data type in Arduino?**	`char` is used to store a **single character** and occupies **one byte**. For example: `char c = 'A';`. Characters are stored using ASCII values.
3	**What is the difference between `int` and `unsigned int`?**	`int` stores values from **−32,768 to 32,767**, whereas `unsigned int` stores only positive values from **0 to 65,535** on an Arduino Uno.
4	**What is the range of the `byte` data type?**	The `byte` data type stores an **8-bit unsigned number** in the range **0 to 255**. Example: `byte m = 25;`.
5	**What is a `float` data type?**	`float` is used to store **decimal or fractional values**. It occupies **32 bits (4 bytes)** and is useful for representing analog or continuous values.
6	**What is variable scope? Name its types.**	Variable scope is the **region of a program where a variable can be accessed**. The three types are **local variables, formal parameters, and global variables**.
7	**What is a local variable?**	A local variable is declared **inside a function or block** and can be used only by the statements within that function or block.
8	**What is a global variable?**	A global variable is declared **outside all functions**, usually at the top of the program. It can be accessed by different functions throughout the program.
9	**What is the purpose of functions in Arduino programming?**	Functions divide a program into **smaller sections that perform individual tasks**. They make programs organized, reusable, compact, and easier to debug.
10	**Name the two required functions in an Arduino sketch.**	The two required functions are **`setup()`** and **`loop()`**. The `setup()` function runs for

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		initialization, while `loop()` contains statements that are repeatedly executed.
11	**What is the use of the `delay()` function?**	The `delay()` function pauses program execution for a specified time in **milliseconds**. Its syntax is `delay(ms);`.
12	**What is the difference between `delay()` and `delayMicroseconds()`?**	`delay()` specifies a delay in **milliseconds**, whereas `delayMicroseconds()` specifies a delay in **microseconds**. For longer delays, `delay()` should be used.
13	**What is the purpose of the `millis()` function?**	`millis()` returns the **number of milliseconds elapsed since the Arduino program started**. The value overflows and returns to zero after approximately **50 days**.
14	**What is an array in Arduino programming?**	An array stores **multiple values of the same data type**. Array elements are accessed using an index, and indexing starts from **0**.
15	**What is an Arduino String? Name any two String functions.**	An Arduino `String` object is used for **storing and manipulating text**. Two useful String functions are **`length()`** and **`concat()`**. Other functions include `indexOf()` and `substring()`.